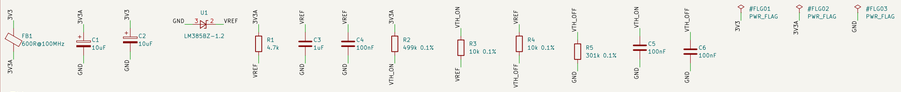
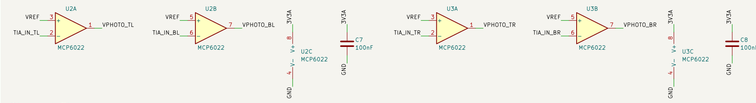
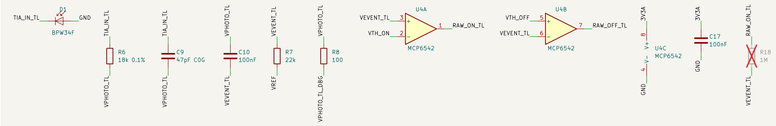
Power & Reference



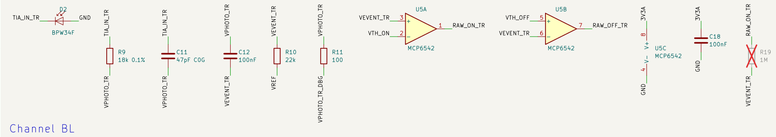
TIA



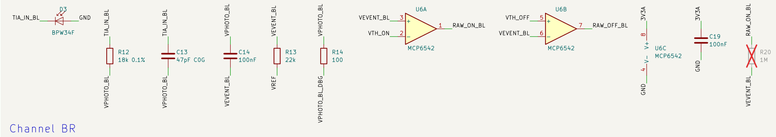
Channel TL



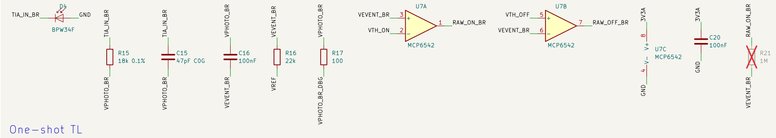
Channel TR



Channel BL

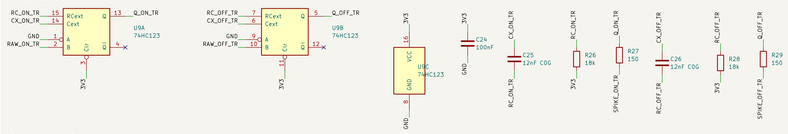


Channel BR

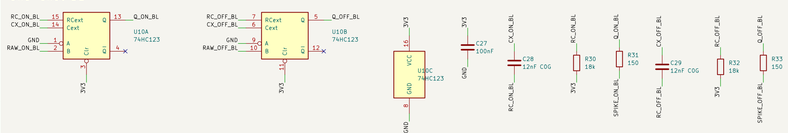


One-shot TL

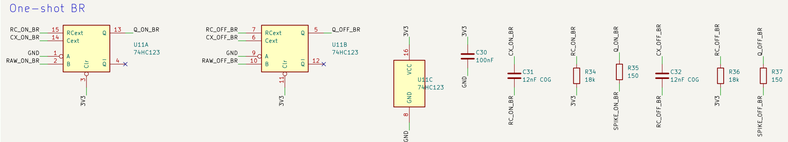
One-shot TR



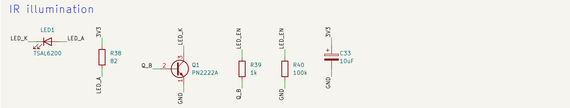
One-shot BL



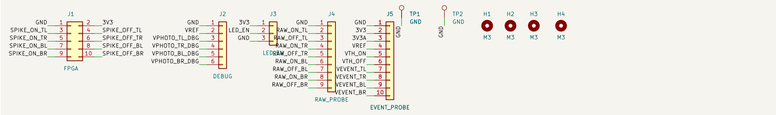
One-shot BR



IR illumination



Connectors



Neuromorphic IR Event Sensor — Rev A-THT (all through-hole, hand-assemblable).

Single source of truth for the netlist. gen_schematic.py turns this into the
KiCad schematic; the PCB is then populated from the schematic netlist.

Deviations from docs/DESIGN_SPECIFICATION.md (SMD Rev A), forced by THT availability:

OPA381 (MSOP)	-> MCP6022 dual CMOS RRIO op-amp, DIP-8 (2 pcs)
TLV3202 (SOIC)	-> MCP6542 dual push-pull comparator, DIP-8 (4 pcs)
74LVC1G123 x8	-> 74HC123 dual retriggerable monostable, DIP-16 (4 pcs)
REF3312 1.25 V	-> LM385BZ-1.2 1.235 V shunt reference, TO-92
AQ3400A	-> PN2222A NPN, TO-92, 1k base resistor
VBPW34FASR	-> OSRAM BPW 34 F (THT DIL, daylight filter)
0603/1206 passives	-> axial resistors, radial ceramic/electrolytic capacitors

All through-hole, hand-assemblable variant

Sheet: /
File: neuromorphic-ir-sensor-tht.kicad_sch

Title: Neuromorphic IR Event Sensor — Rev A-THT

Size: A1	Date:	**Rev: A-THT**
KiCad E.D.A. 10.0.6		Id: 1/1